

Software Engineering

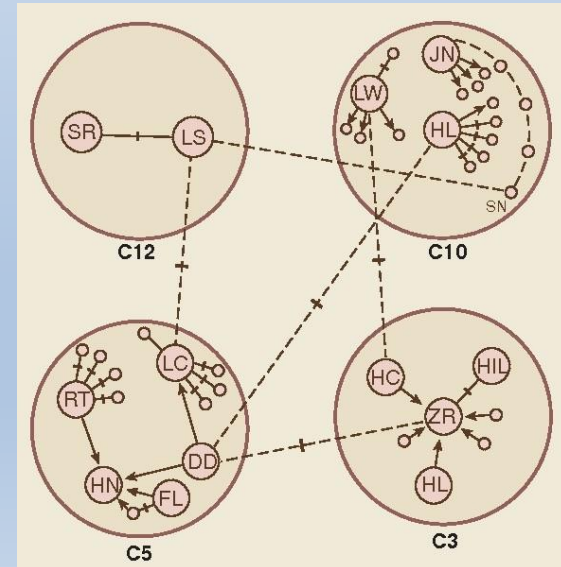
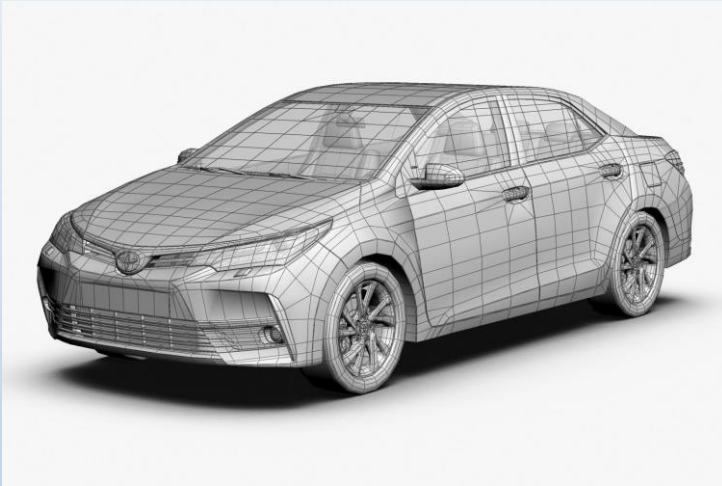
Fall 2026

LECTURE 6

Part I

MODELS

Models



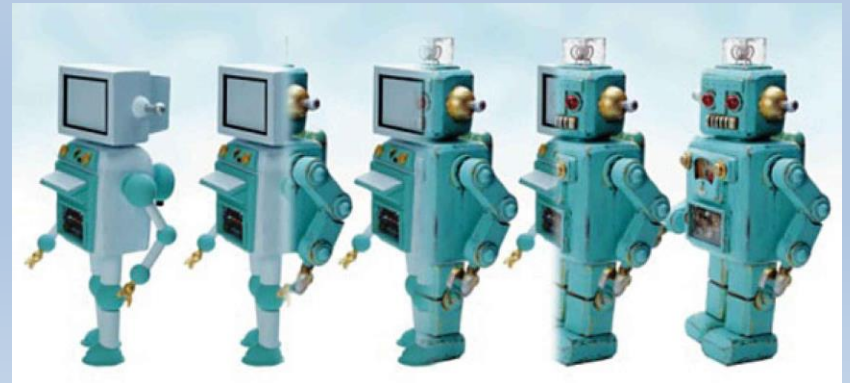
What represent Models?

- **Abstraction**
 - Remove detail
 - Focus in viewpoint
- **Dynamic**
 - Interaction with environment
 - Control flow
 - Data flow
- **Static**
 - Physical
 - Database schema
 - Object structure

Prototypes

- **Used for**
 - Requirements elicitation
 - Requirements validation
 - Proof of concept
 - User training
 - Back to back testing

Risk reduction!



Be careful with Prototypes

- **A prototype is not a product!** *(it might differ in)*
 - Functionality
 - Performance
 - Reliability
 - Scalability
 - Maintainability
 - User interface

A model is not a prototype

- **Model**
 - Used to understand some aspect of a system
- **Prototype**
 - Fully functional "first try" of a system

Prototypes are models, because they help us understand how a system will perform. But most models are not prototypes.

System modeling

- Process of developing abstract models of a system, with each model presenting a different view or perspective of that system.
- Representing a system using some kind of graphical notation, which is now almost always based on notations in the Unified Modeling Language (UML).
- Helps the analyst to understand the functionality of the system and models are used to communicate with customers.

Existing and planned models

- Models of the existing system are used during requirements engineering. They help clarify what the existing system does and can be used as a basis for discussing its strengths and weaknesses. These then lead to requirements for the new system.
- Models of the new system are used during requirements engineering to help explain the proposed requirements to other system stakeholders. Engineers use these models to discuss design proposals and to document the system for implementation.
- In a model-driven engineering process, it is possible to generate a complete or partial system implementation from the system model.

Graphical Models

- As a means of **facilitating discussion** about an existing or proposed system
 - Incomplete and incorrect models are OK as their role is to support discussion.
- As a way of **documenting an existing** system
 - Models should be an accurate representation of the system but need not be complete.
- As a **detailed system description** that can be used to generate a **system implementation**
 - Models have to be both correct and complete.

Model design in SE

- UML (*Unified Modeling Language*)

Wikipedia: The **Unified Modeling Language (UML)** is a general-purpose, developmental, **modeling language** in the field of **software engineering**, that is intended to provide a standard way to visualize the design of a system.

- Warnings:
 - *Try to avoid design*
 - *Avoid massive documentation*

Use in Project!

Part II

UML

- **An external perspective**

- Model the context or environment of the system.

- **An interaction perspective**

- Model the interactions between a system and its environment, or between the components of a system.

- **A structural perspective**

- Model the organization of a system or the structure of the data that is processed by the system.

- **A behavioral perspective**

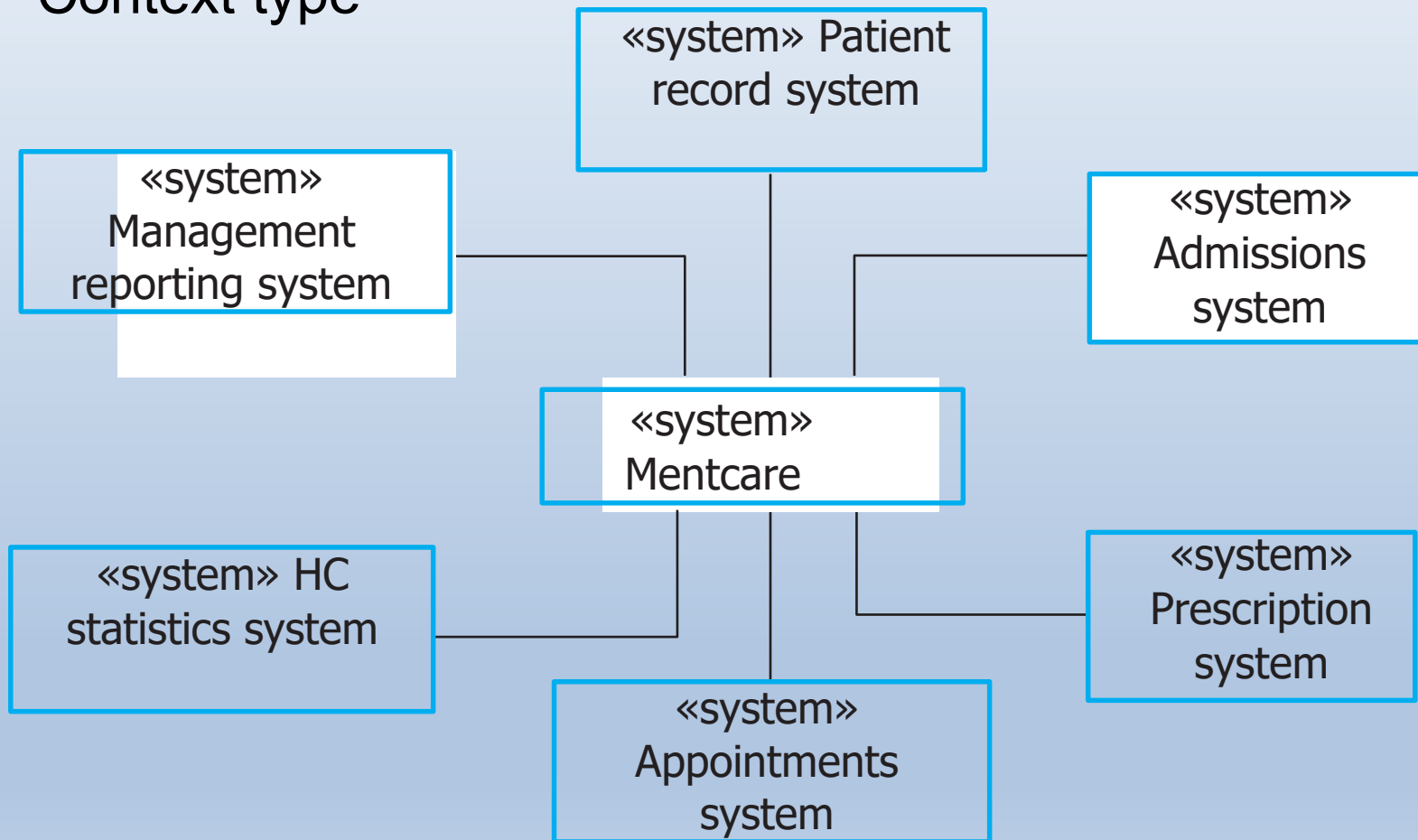
- Model the dynamic behavior of the system and how it responds to events.

Context Models

- Context models are used to illustrate the operational context of a system - they show what lies outside the system boundaries.
- Social and organisational concerns may affect the decision on where to position system boundaries.
- Architectural models show the system and its relationship with other systems.

UML (simplified) Component Diagram

- Context type



UML (simplified) Context Diagram

- UML Context diagram for Coukoo Clock

1.PlantUML: uses a simple text syntax to create UML diagrams can be integrated into various environments.

2.Lucidchart: also offers a free version with limited functionality

3.Draw.io: a free online diagram tool with robust support for creating UML diagrams.

4.StarUML: free version provides many features for basic UML modelling.

5.ArgoUML: easy to use and runs on many platforms.

6.Modelio: A free modelling environment that supports UML and other modelling standards like BPMN.

7.UMLet: A free UML tool with a simple, user-friendly interface.

Interaction Models

- Modeling **user interaction**
 - Important as it helps to identify user requirements.
- Modeling **system-to-system interaction**
 - Highlights the communication problems that may arise.
- Modeling **component interaction**
 - Helps to understand system structure is able to deliver the required system performance and dependability.
- **Use case** diagrams and sequence diagrams may be used for interaction modeling.

UML Use Case Modeling

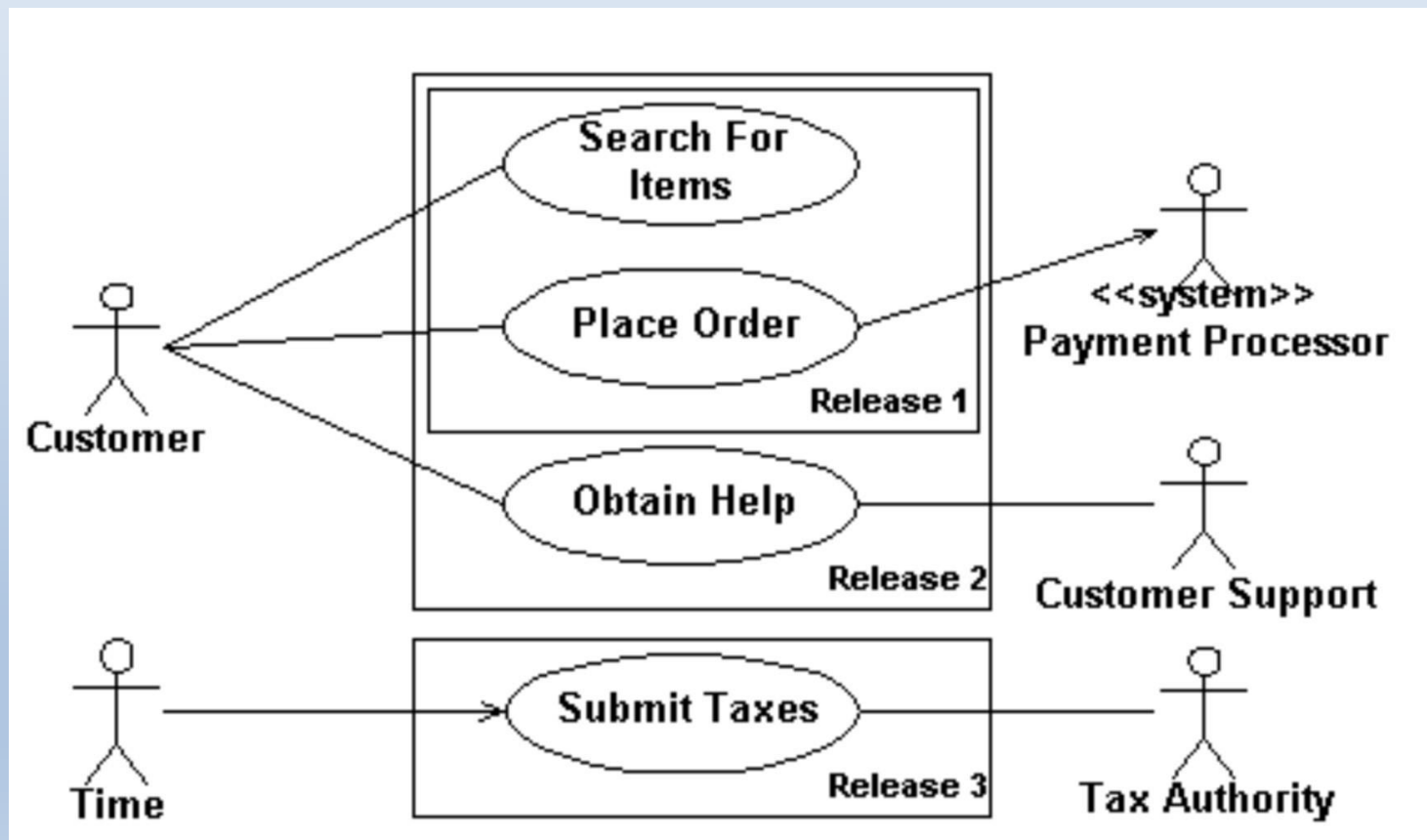
- Use cases were developed originally to support requirements elicitation and now incorporated into the UML.
- Each use case represents a discrete task that involves external interaction with a system.
- Actors in a use case may be people or other systems.
- Represented diagrammatically to provide an overview of the use case and in a more detailed textual form.

UML Use Case Modeling

UML Use Case diagram for Coukoo Clock

UML Use Case Diagram

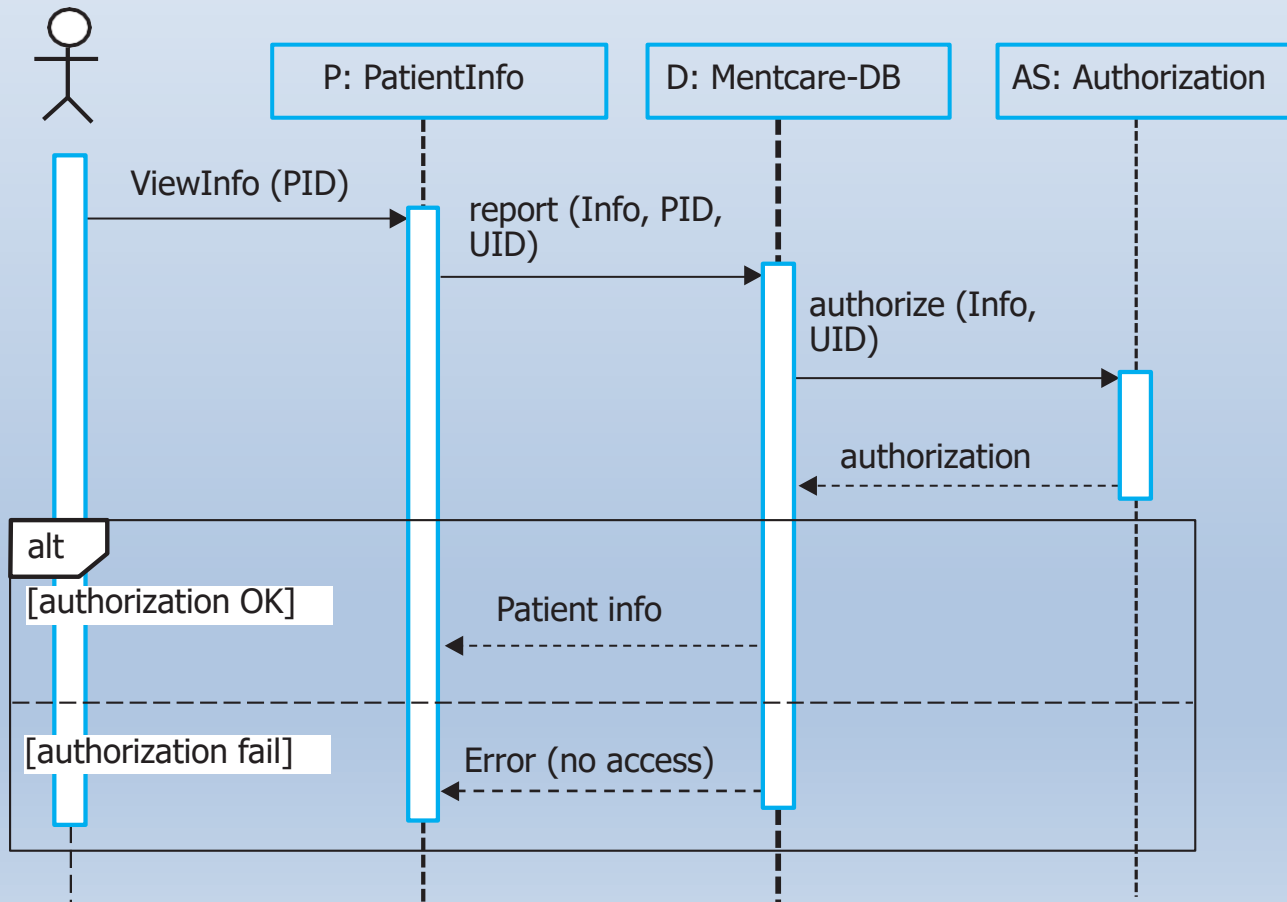
- Interaction type



UML Sequence Diagram

- Interaction type

Medical Receptionist

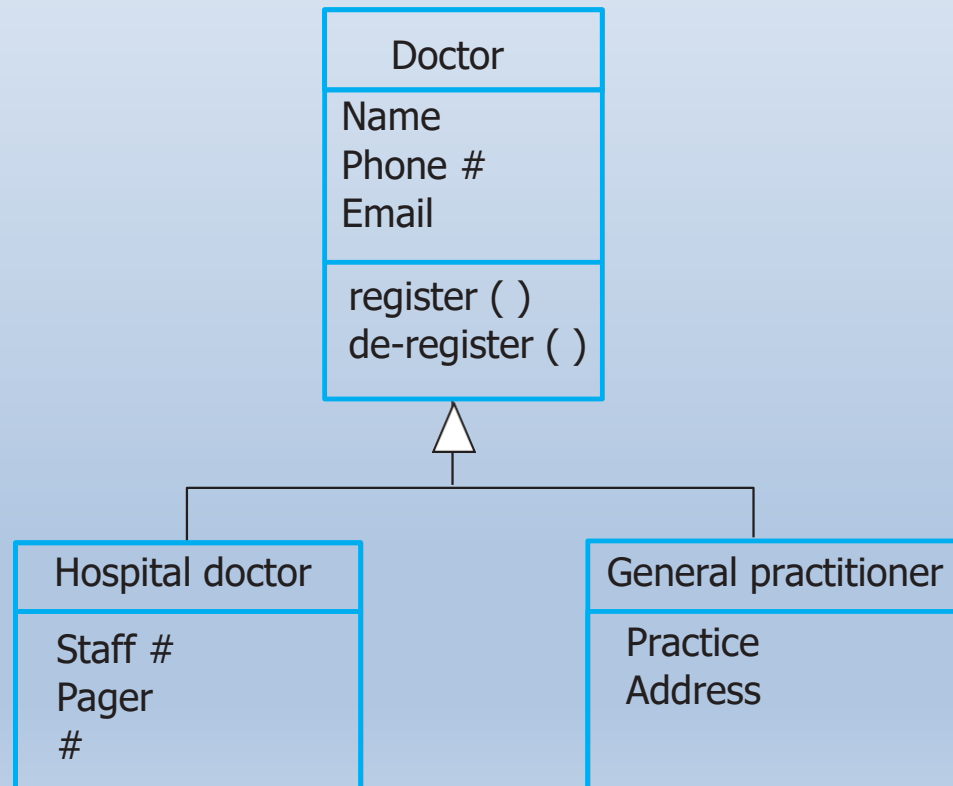


UML Sequence Diagram

UML Sequence diagram
for
Coukoo Clock

UML Class Diagram

- Structure type
 - *Check example in course material!*

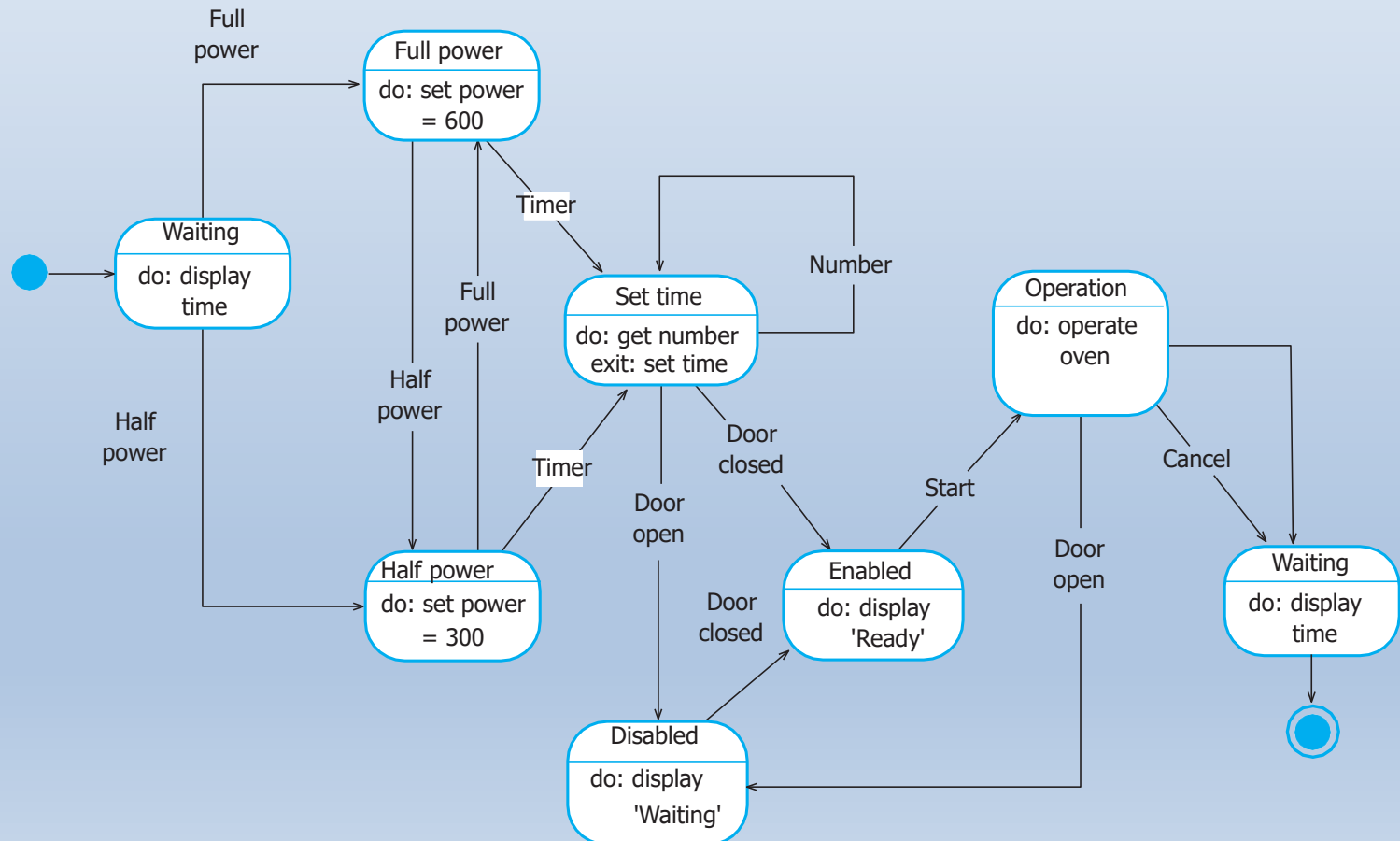


UML Sequence Diagram

UML diagram
for
Coukoo Clock

UML State Diagram

- Behaviour type



UML State Diagram

```
stateDiagram StateDiagramCuckooClock
    [*] --> Idle
    Idle --> Cuckooing : Hour Change
    Cuckooing --> Cuckooing_Pause : Cuckooing Complete
    Cuckooing_Pause --> Idle : Cuckooing Retraction Complete
    Idle --> Winding : Winding Key Turned
    Winding --> Idle : Winding Key Released
```

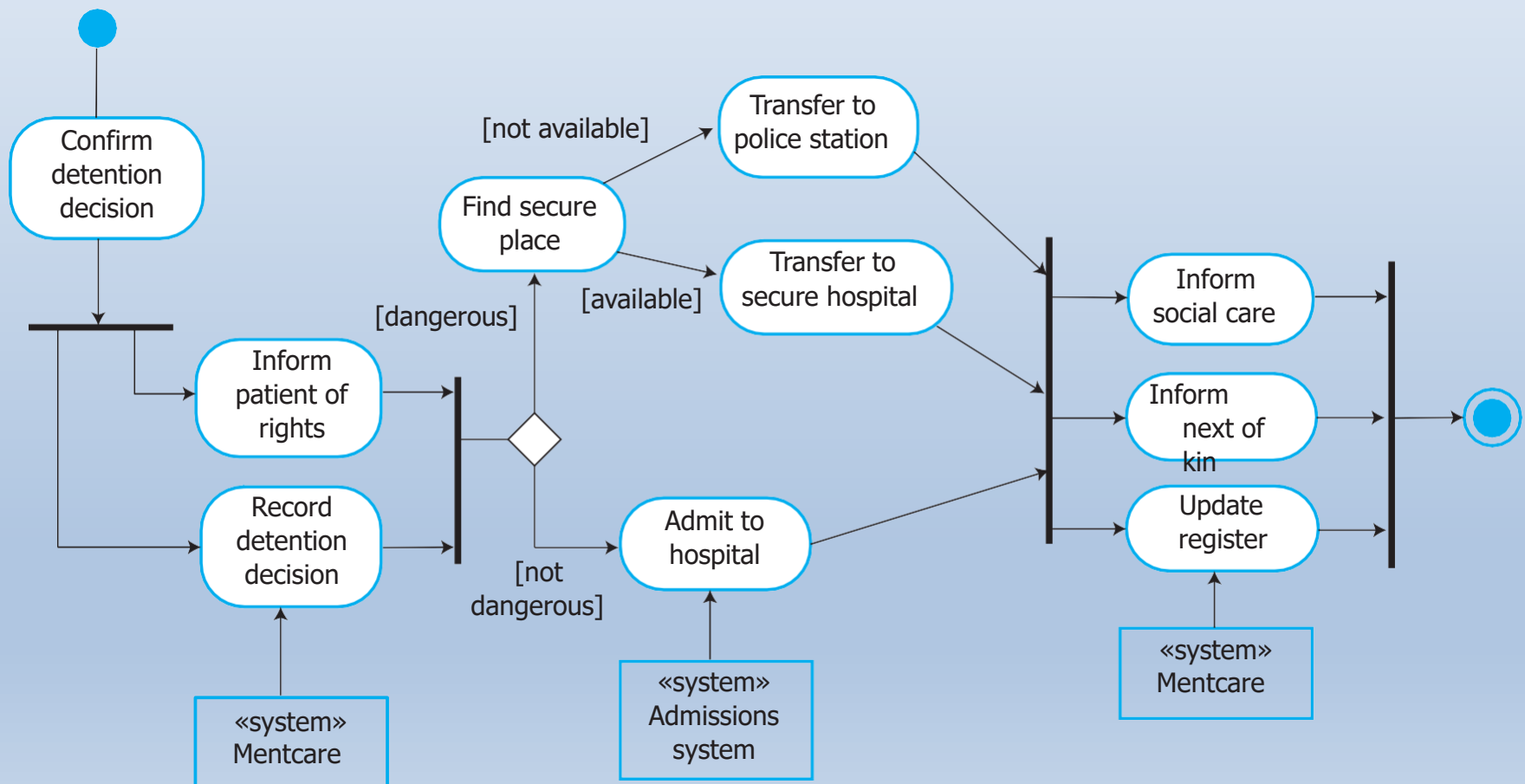
```
state Cuckooing {
    [*] --> Sing Cuckoo
}
```

```
state Cuckooing_Pause {
    [*] --> Cuckoo Bird Retraction
}
```

```
state Winding {
    [*] --> Wind Mechanism
}
```

UML Activity Diagram

- Behaviour type



UML (Behaviour type)

- State vs. Activity Diagrams
- **state diagrams** are about the states and transitions of objects.
 - states, transitions, events, actions
- **activity diagrams** focus on the flow of activities and actions in a system.
 - activities, transitions, decisions, swimlanes, forks, and joins

References

	9ed	10ed
Chapters	5, 2.3 See 10ed Chap. 3	5. Intro 5.2, 5.4, 2.3.1, 2.3.2, 3
Important Sections	5-intro, 5.2, 5.4, 2.3.1, 2.3.1, 2.3.2 See 10ed Chap. 3.4	5. Intro. Systems modeling 5.2 interaction models, 5.4 Behavioral models, 2.3.1 Prototyping, 2.3.2 Incremental delivery

Questions

